

M&E; Dashboard — DHQ Hospitals Revamping

Python Data Analysis Project | Python • Pandas • NumPy • Matplotlib • Seaborn

Project Overview

Performed end-to-end data analysis for the **DHQ Hospitals Revamping Project** using Python. The analysis focuses on project performance, targets and achievements, activity status, district-wise performance, budget allocation and utilization, and the relationship between budget utilization and achievement.

Tools & Libraries

Tool / Library	Use in the Project
Python	Core data analysis workflow
Pandas	Excel loading, data exploration, grouping, aggregation, cleaning and analysis
NumPy	Numerical/data analysis support
Matplotlib	Chart creation and visualization
Seaborn	Data visualization library imported for the analysis

Dataset Loading & Exploration

- Loaded the Excel workbook using **pd.ExcelFile()** and checked available sheets.
- Loaded the **Python Data** sheet using **pd.read_excel()**.
- Inspected columns, first and last rows, dataset shape, data types, and descriptive statistics.

Data Quality & Cleaning

- Checked missing values for each column and calculated total missing values.
- Checked for duplicate rows using **df.duplicated().sum()**.
- Removed duplicate rows using **df.drop_duplicates()**, keeping the first occurrence.
- Verified the dataset shape after duplicate removal.

Performance Analysis

- Calculated **category-wise achievement** using groupby aggregation.
- Calculated **quarter-wise achievement** to analyze performance trends.
- Calculated **activity status counts** for Completed, Pending, and other statuses present in the data.
- Calculated district-wise **Target** and **Achievement** totals.
- Calculated **Achievement Percentage = Achievement / Target × 100**.
- Identified the highest and lowest district achievement percentages using **idxmax()** and **idxmin()**.

Budget Analysis

- Calculated total **Budget Allocated (PKR)** and total **Budget Utilized (PKR)**.
- Performed district-wise budget analysis for allocated and utilized amounts.
- Calculated **Budget Utilization Percentage = Utilized / Allocated × 100**.
- Sorted district budget utilization percentages for comparison.

Correlation Analysis

Calculated the correlation between **Budget Utilized** and **Achievement** using Pandas `.corr()` to examine their linear relationship.

Visualizations Created

Chart	Analysis
1. District-wise Achievement	Total achievement by district
2. Category-wise Achievement	Total achievement by category
3. Quarterly Achievement Trend	Achievement trend by quarter
4. Budget Allocated vs Utilized	Comparison of allocated and utilized budget
5. Activity Status Distribution	Number of activities by status
6. District-wise Budget Utilization	Budget utilization percentage by district

Chart Generation Workflow

A reusable `plot_and_save()` function was created to generate the charts. The function supports bar and line charts, applies titles and axis labels, rotates category labels where required, saves each visualization as a PNG file, and displays the chart.

Project Outputs

- Six Python-generated PNG visualizations.
- District-wise, category-wise, quarterly, status, and budget analysis.
- Achievement percentage and budget utilization percentage calculations.
- Correlation analysis between budget utilized and achievement.

Project Skills Demonstrated

Python Data Analysis • Pandas • NumPy • Data Cleaning • Exploratory Data Analysis • GroupBy & Aggregation • Performance Analysis • Budget Analysis • Correlation Analysis • Matplotlib • Seaborn • Data Visualization • PNG Chart Generation

Project Category

02 / PYTHON DATA ANALYSIS